

From four levels to the Dicke model

the adiabatic elimination, done explicitly

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Abstract

A companion note reduces the four-level cavity Rydberg-EIT problem all the way to a Dicke model. Here we stop two steps earlier, at the three-level cavity Hamiltonian $H^{(3)}$ of Eq. (13), and spend the space instead on Step 3 — the elimination of the intermediate state $|2\rangle$, which is where the Raman coupling $\lambda = g_0\Omega_c/2\Delta$, the light shifts and the residual spontaneous emission all come from. One line of second-order perturbation theory does it, $H_{mn}^{\text{eff}} = \tilde{H}_{mn} - V_m V_n^\dagger / \Delta$, and that one line pays for itself three times: its rank-one structure shows that λ and the dominant systematic cannot be tuned independently; putting Γ_e in the denominator returns the inherited decay rates and the invariant $\lambda^2/\gamma_3 = g_0^2/\Gamma_e$, which says neither the detuning nor the control power improves the sensor; and keeping the next order returns the error of the reduction, verified against exact diagonalisation with no fitted constant. Throughout $\hbar = 1$.

1 The exact Hamiltonian

Four atomic levels with laboratory energies $\omega_1 = 0 < \omega_2 < \omega_3 < \omega_4$:

$$|1\rangle = |5S_{1/2}\rangle, \quad |2\rangle = |5P_{3/2}\rangle, \quad |3\rangle = |53D_{5/2}\rangle, \quad |4\rangle = |54P_{3/2}\rangle,$$

numerically $\omega_2/2\pi = 384.25$, $\omega_3/2\pi = 1008.82$ THz, and $(\omega_4 - \omega_3)/2\pi \simeq 10$ GHz. Writing $\sigma_{jk} = |j\rangle\langle k|$, and with no approximations whatsoever,

$$\begin{aligned} H = & \underbrace{\sum_{j=1}^4 \omega_j \sigma_{jj} + \omega_{\text{cav}} a^\dagger a}_{\text{free}} + \underbrace{g_0 (a + a^\dagger)(\sigma_{12} + \sigma_{21})}_{\text{cavity, } |1\rangle \leftrightarrow |2\rangle} \\ & + \underbrace{\Omega_c \cos(\nu_c t) (\sigma_{23} + \sigma_{32})}_{\text{control laser}} + \underbrace{\Omega_\mu \cos(\nu_\mu t) (\sigma_{34} + \sigma_{43})}_{\text{microwave}}. \end{aligned} \quad (1)$$

The classical fields carry their full time dependence; the cavity coupling carries both a and $a^\dagger$. Nothing has been dropped.

2 Step 1: the rotating frame

Transform with $\tilde{H} = U H U^\dagger + i\dot{U}U^\dagger$, choosing

$$U(t) = \exp\left\{it[\omega_L a^\dagger a + \omega_L \sigma_{22} + (\omega_L + \nu_c) \sigma_{33} + (\omega_L + \nu_c + \nu_\mu) \sigma_{44}]\right\}, \quad (2)$$

i.e. each level is rotated at the *sum of the drive frequencies needed to reach it* (ω_L is a reference near the cavity resonance). This is an exact change of picture. Its whole effect is on the diagonal:

level	rotated at	lab energy	rotating-frame energy
$ 1\rangle$	0	0	0
$ 2\rangle$	ω_L	$2\pi \times 384$ THz	$\Delta \equiv \omega_2 - \omega_L$ ($\sim$ GHz)
$ 3\rangle$	$\omega_L + \nu_c$	$2\pi \times 1009$ THz	$\delta \equiv \omega_3 - \omega_L - \nu_c$ ($\sim$ MHz)
$ 4\rangle$	$\omega_L + \nu_c + \nu_\mu$	$\omega_3 + 2\pi \times 10$ GHz	$\delta + \Delta_\mu$, $\Delta_\mu \equiv (\omega_4 - \omega_3) - \nu_\mu$

This answers the obvious objection. $|3\rangle$ and $|4\rangle$ are *not* degenerate — they are 10 GHz apart. But the frame rotates $|4\rangle$ faster than $|3\rangle$ by exactly ν_μ , and ν_μ is chosen close to that splitting. The 10 GHz is subtracted off; what remains on the diagonal is only the residual *microwave detuning* Δ_μ , of order MHz. Setting $\Delta_\mu = 0$ makes the two entries equal — and that is a statement about the *drive*, not about the atom.

3 Step 2: the rotating-wave approximation

In the new frame the cavity term $a^\dagger \sigma_{21}$ picks up $e^{2i\omega_L t}$, and $\cos(\nu t) \sigma_{jk}$ splits into a static piece and one at $e^{2i\nu t}$. Dropping the fast pieces requires

$$g_0, \Omega_c, \Omega_\mu \ll \omega_L, \nu_c, \nu_\mu, \quad (\text{A1})$$

which is spectacularly well satisfied (MHz against THz, and MHz against the 10 GHz microwave). The result is time-independent and still exact to this order:

$$\tilde{H} = \omega_c a^\dagger a + \begin{pmatrix} 0 & g_0 a^\dagger & 0 & 0 \\ g_0 a & \Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \omega_c \equiv \omega_{\text{cav}} - \omega_L. \quad (3)$$

Note the matrix is tridiagonal: each field connects only neighbouring levels.

4 Step 3: eliminate the intermediate state

$|2\rangle$ sits $\Delta \sim 2\pi \times 100$ MHz away from anything that drives it, so it is never appreciably populated; its only job is to relay. That is precisely the situation second-order perturbation theory was built for, and one formula does the whole step.

4.1 One formula

List the couplings that reach $|2\rangle$ from the states we keep. Reading the second column of Eq. (3),

$$V_m \equiv \langle m | \tilde{H} | 2 \rangle : \quad V_1 = g_0 a^\dagger, \quad V_3 = \Omega_c/2, \quad V_4 = 0. \quad (4)$$

The V_m are operators on the cavity, not numbers, and $V_4 = 0$ says the microwave never touches $|2\rangle$. Quasi-degenerate second-order perturbation theory then gives the effective Hamiltonian on $\{|1\rangle, |3\rangle, |4\rangle\}$ as

$$H_{mn}^{\text{eff}} = \tilde{H}_{mn} - \frac{V_m V_n^\dagger}{\Delta} \quad m, n \in \{1, 3, 4\}, \quad (5)$$

one term per *path* $m \rightarrow 2 \rightarrow n$: two matrix elements over one energy denominator. (The denominator is properly $\frac{1}{2}[(E_m - \Delta)^{-1} + (E_n - \Delta)^{-1}]$, but $E_{1,3,4} = 0, \delta, \delta + \Delta_\mu$ are all $\sim$ MHz against $\Delta \sim 100$ MHz, so $-1/\Delta$ it is; the δ/Δ correction is smaller than the error we quantify in §4.5.)

Four paths are non-zero, and evaluating them is the entire step:

path	$V_m V_n^\dagger$	contributes	is
$1 \rightarrow 2 \rightarrow 1$	$g_0 a^\dagger \cdot g_0 a = g_0^2 a^\dagger a$	$-g_0^2 a^\dagger a / \Delta$	a cavity shift
$3 \rightarrow 2 \rightarrow 3$	$\Omega_c^2 / 4$	$-\Omega_c^2 / 4\Delta$	a light shift on $ 3\rangle$
$1 \rightarrow 2 \rightarrow 3$	$g_0 a^\dagger \cdot \Omega_c / 2$	$-\lambda a^\dagger$	the Raman coupling
$3 \rightarrow 2 \rightarrow 1$	$\Omega_c / 2 \cdot g_0 a$	$-\lambda a$	its conjugate

Anything involving $|4\rangle$ vanishes because $V_4 = 0$. Assembling them,

$$\tilde{H}^{(3)} = \omega_c a^\dagger a + \begin{pmatrix} -\frac{g_0^2 a^\dagger a}{\Delta} & -\lambda a^\dagger & 0 \\ -\lambda a & \delta - \frac{\Omega_c^2}{4\Delta} & \Omega_\mu / 2 \\ 0 & \Omega_\mu / 2 & \delta + \Delta_\mu \end{pmatrix}, \quad \boxed{\lambda = \frac{g_0 \Omega_c}{2\Delta}} \quad (6)$$

That is Step 3. Two features of Eq. (5) are worth pausing on, because both have consequences later. It is an *outer product*, $V_m V_n^\dagger$, hence rank one — the two light shifts and the Raman coupling are three views of one object, not three knobs. And the operator ordering came out right on its own: $V_1 V_1^\dagger = g_0 a^\dagger g_0 a \propto a^\dagger a$, normally ordered, so an empty cavity receives no shift. Guessing the answer rather than computing it risks an $n+1$ where the truth is n .

4.2 This is the rotating-wave approximation again

It is worth saying plainly, because it makes Step 3 an application of Step 2 rather than a new idea. In both steps something oscillates fast compared with everything driving it: in Step 2 the counter-rotating field at 2ν , here the amplitude of $|2\rangle$ at Δ . In both, the fast thing averages to zero at first order and leaves a second-order residue of size (amplitude)²/(frequency) — which is exactly what Eq. (5) computes, with V the amplitude and Δ the frequency. The validity conditions (A1) and (A2) are the same statement twice. Appendix A does the same elimination on the amplitude equations for anyone who wants to watch $|2\rangle$ become a shadow of the states that drive it.

Equation (5) is also the leading term of something exact. Keeping the energy E of the state one is after, rather than setting it to zero, the reduction $H_{mn}^{\text{eff}}(E) = \tilde{H}_{mn} + V_m(E - \Delta)^{-1} V_n^\dagger$ is exact to all orders; expanding $(E - \Delta)^{-1}$ in E/Δ returns Eq. (5) at leading order and, at the next one, the error quantified in §4.5.

4.3 Reading the three terms

$-\lambda(a^\dagger \sigma_{13} + a \sigma_{31})$, **the Raman coupling**. A cavity photon is absorbed on $|1\rangle \rightarrow |2\rangle$ and a control photon emitted on $|2\rangle \rightarrow |3\rangle$, with $|2\rangle$ visited only virtually. It is the term the whole scheme is built on, and it is *first* order in each field: halving the control power halves λ . The minus sign is a phase convention ($|3\rangle \rightarrow -|3\rangle$ removes it) and we drop it.

$-\Omega_c^2/4\Delta$ on $|3\rangle$, **the control light shift**. Absorbed by redefining $\delta \rightarrow \delta - \Omega_c^2/4\Delta$. Harmless as bookkeeping, not harmless as physics — see the box at the end of this section.

$-(g_0^2/\Delta) a^\dagger a$ on $|1\rangle$, **the dispersive shift**. This multiplies $a^\dagger a$, so it is not an atomic energy at all but a shift of the *cavity*, and with N atoms in $|1\rangle$ it adds up: $\omega_c \rightarrow \omega_c - N g_0^2/\Delta$. For $N = 10^6$, $g_0/2\pi = 1$ MHz, $\Delta/2\pi = 100$ MHz that is $2\pi \times 10$ GHz against $\omega_c/2\pi \sim 1$ MHz. It is not a perturbation to be dropped but a redefinition of where the cavity sits, absorbed in the laboratory by locking to the shifted resonance. What matters for the sensor is that it drifts with N and with Δ .

4.4 Keeping the loss

$|2\rangle$ radiates at $\Gamma_e/2\pi \simeq 6$ MHz. Nothing above changes if we let the denominator in Eq. (5) go complex, $\Delta \rightarrow \Delta - i\Gamma_e/2$, the standard bookkeeping for a level whose population leaves the system. Expanding $(\Delta - i\Gamma_e/2)^{-1} \simeq \Delta^{-1} + i\Gamma_e/2\Delta^2$, the two diagonal paths acquire imaginary parts:

$$\gamma_3 = \frac{\Omega_c^2}{4\Delta^2} \Gamma_e \quad (\text{Rydberg state}), \quad \kappa_{\text{abs}} = \frac{g_0^2}{\Delta^2} \Gamma_e \quad (\text{cavity}) \quad (7)$$

Each is Γ_e times the residual $|2\rangle$ population the corresponding drive maintains, $(\Omega_c/2\Delta)^2$ and $(g_0/\Delta)^2$ per intracavity photon: the elimination has not removed the decay, only diluted it. At $\Omega_c/2\pi = 5$, $g_0/2\pi = 1$, $\Delta/2\pi = 100$ MHz these are $2\pi \times 3.75$ and 0.60 kHz, confirmed against the exact eigenvalues to better than 0.5%.

The scaling looks at first like a free lunch — $\lambda \propto \Delta^{-1}$ but $\gamma_3 \propto \Delta^{-2}$, so detuning further seems to buy coherence faster than it costs coupling. It does not. Form the combination in which Δ and Ω_c both cancel:

$$\frac{\lambda^2}{\gamma_3} = \frac{g_0^2 \Omega_c^2 / 4\Delta^2}{\Omega_c^2 \Gamma_e / 4\Delta^2} = \frac{g_0^2}{\Gamma_e} \quad (8)$$

independent of *both* knobs Step 3 gives us. Detuning further and raising the control power to compensate slides λ and γ_3 along a line of constant λ^2/γ_3 . Divide by the cavity linewidth and, with $\lambda \rightarrow \lambda\sqrt{N}$ at fixed single-atom γ_3 , this is the familiar figure of merit and the only one in the problem, $N\lambda^2/\kappa\gamma_3 = Ng_0^2/\kappa\Gamma_e = NC$. Atom number and cavity finesse improve the sensor; Δ and Ω_c only decide where on the line one sits. The cavity absorption is the exception that proves the point — $\lambda^2/\kappa_{\text{abs}} = \Omega_c^2/4\Gamma_e$ *does* reward control power, because κ_{abs} has no Ω_c in it.

4.5 How good is it, and when is it allowed?

Restoring the energy dependence of the resolvent, $(E - \Delta)^{-1} = -\Delta^{-1}(1 + E/\Delta + \dots)$, gives the leading error directly:

$$\delta E \simeq \frac{E \langle VV^\dagger \rangle}{\Delta^2}, \quad \langle VV^\dagger \rangle = g_0^2 \langle a^\dagger a \rangle P_1 + \frac{\Omega_c^2}{4} P_3 \quad (9)$$

with P_1, P_3 the populations of $|1\rangle, |3\rangle$ in the state whose energy is E . The error is second order in $1/\Delta$ while λ is only first order, so the model gets *relatively* better as Δ grows even though the physics it describes gets weaker; and a state with no $|3\rangle$ component is described far more accurately than the dressed states, since only g_0 , not Ω_c , reaches $|2\rangle$ from $|1\rangle$.

Figure 1 tests both statements. Equation (3) conserves $N_{\text{exc}} = a^\dagger a + \sigma_{22} + \sigma_{33} + \sigma_{44}$, so its single-excitation sector ($|1, n=1\rangle, |2, 0\rangle, |3, 0\rangle, |4, 0\rangle$) is a closed 4×4 problem and the comparison needs no truncation.

Collecting the conditions,

$$|\Delta| \gg \Omega_c, \quad g_0 \sqrt{\langle a^\dagger a \rangle}, \quad \Omega_\mu, \quad |\delta|, \quad |\omega_c|, \quad \Gamma_e. \quad (\text{A2})$$

The first two are the amplitudes reaching $|2\rangle$, the next three the rates at which they turn over, the last the width of the state being eliminated. It is the *full* list: a treatment that checks only $|\Delta| \gg \Omega_c, \Gamma_e$ is caught out at high intracavity photon number, where $g_0 \sqrt{\langle n \rangle}$ creeps up on Δ . At $\Delta/2\pi = 100$ and $g_0/2\pi = 1$ MHz the elimination survives to $\langle n \rangle \sim 10^3$ and no further.

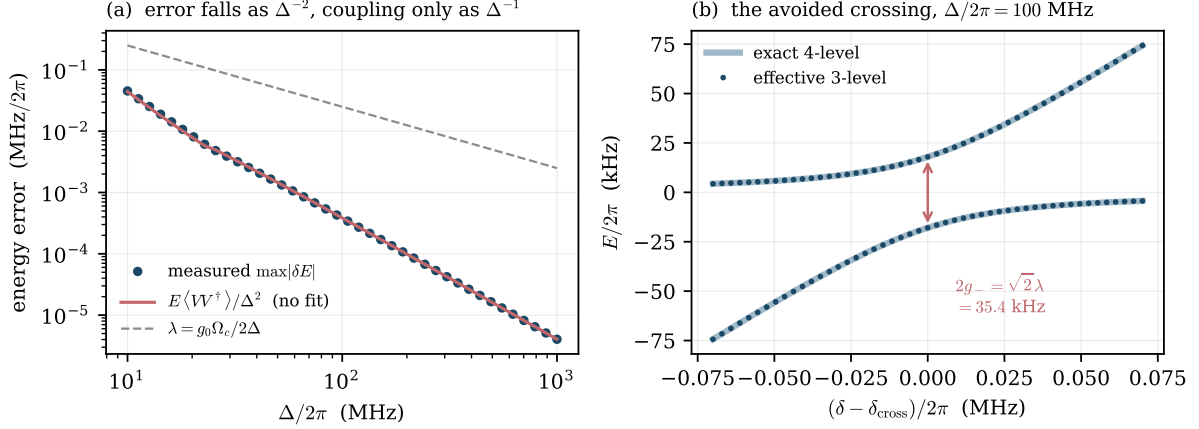


Figure 1: **Step 3 tested against exact diagonalisation**, in the single-excitation sector at $g_0/2\pi = 1$, $\Omega_c/2\pi = 5$, $\Omega_\mu/2\pi = 2$, $\delta/2\pi = 0.3$ MHz, $\Delta_\mu = 0$. (a) Worst-case eigenvalue error versus Δ (points) against Eq. (9) with no fitted parameter (line); the Raman coupling λ is shown for scale, falling only as Δ^{-1} . (b) The avoided crossing at $\Delta/2\pi = 100$ MHz. Measured minimum gap 35.9 kHz against the predicted $2g_- = \sqrt{2}\lambda = 35.4$ kHz; the 1.5% excess is the $\Omega_c^2/4\Delta$ light shift, which sits on $|3\rangle$ alone and so slightly reshapes the dressing of Step 4.

Where the dominant systematic comes from. $V_m V_n^\dagger$ has rank one, so the Raman coupling and the light shift are not independent knobs: $\lambda \propto \Omega_c$ and the shift $\propto \Omega_c^2$ both track the control-laser intensity, and the shift adds *directly* to δ — the same channel the measurand uses. Since $\Omega_c^2 \propto I$, a fractional intensity fluctuation ε moves δ by $\varepsilon \Omega_c^2/4\Delta$, at our numbers $\varepsilon \times 2\pi \times 62$ kHz. Control-laser intensity noise is therefore structurally indistinguishable from signal, and no amount of care with the microwave will fix it.

5 Step 4: dress the Rydberg pair

The lower-right block of Eq. (6) is a two-level problem on its own:

$$H_\mu = \begin{pmatrix} \delta & \Omega_\mu/2 \\ \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix} = \left(\delta + \frac{\Delta_\mu}{2}\right) \mathbf{1} + \frac{1}{2}(\Omega_\mu \sigma_x + \Delta_\mu \sigma_z). \quad (10)$$

Diagonalise with the rotation $R(\theta)$ defined by

$$\tan \theta = \frac{\Omega_\mu}{\Delta_\mu}, \quad |+\rangle = \sin \frac{\theta}{2} |3\rangle + \cos \frac{\theta}{2} |4\rangle, \quad |-\rangle = \cos \frac{\theta}{2} |3\rangle - \sin \frac{\theta}{2} |4\rangle, \quad (11)$$

giving the *generalised Rabi* splitting

$$\omega_\pm = \delta + \frac{\Delta_\mu}{2} \pm \frac{W}{2}, \quad W = \sqrt{\Omega_\mu^2 + \Delta_\mu^2} \quad (12)$$

Now rotate the rest of Eq. (6). The Raman coupling connects $|1\rangle$ to the *bare* $|3\rangle = \sin \frac{\theta}{2} |+\rangle + \cos \frac{\theta}{2} |-\rangle$, so it splits between the branches in that ratio. In the basis $(|g\rangle, |+\rangle, |-\rangle)$:

$$H^{(3)} = \begin{pmatrix} 0 & g_+ a^\dagger & g_- a^\dagger \\ g_+ a & \omega_+ & 0 \\ g_- a & 0 & \omega_- \end{pmatrix}, \quad g_+ = \lambda \sin \frac{\theta}{2}, \quad g_- = \lambda \cos \frac{\theta}{2} \quad (13)$$

There is no $|+\rangle \leftrightarrow |-\rangle$ entry: that transition is the microwave one and carries no cavity matrix element.

Equation (13) is where this note stops. It is still excitation-number conserving, so on its own it radiates nothing; turning it into a Dicke model requires counter-rotating terms $a^\dagger b^\dagger + ab$ from a second drive tone, and that, together with the Holstein–Primakoff step and the superradiant threshold, is the subject of the companion note.

A The same elimination by hand

Equation (5) is quicker, but it is worth seeing once where it comes from. Expand the state as $|\Psi\rangle = \sum_j |j\rangle \otimes |\phi_j\rangle$ with $|\phi_j\rangle$ (unnormalised) cavity states. Reading the rows of Eq. (3),

$$i \partial_t |\phi_1\rangle = \omega_c a^\dagger a |\phi_1\rangle + g_0 a^\dagger |\phi_2\rangle, \quad (14)$$

$$i \partial_t |\phi_2\rangle = \omega_c a^\dagger a |\phi_2\rangle + \Delta |\phi_2\rangle + g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle, \quad (15)$$

$$i \partial_t |\phi_3\rangle = \omega_c a^\dagger a |\phi_3\rangle + \delta |\phi_3\rangle + \frac{\Omega_c}{2} |\phi_2\rangle + \frac{\Omega_\mu}{2} |\phi_4\rangle, \quad (16)$$

and a fourth line for $|\phi_4\rangle$ that never involves $|\phi_2\rangle$. Only Eq. (15) carries a large coefficient. Remove the slow diagonal terms with $|\phi_j\rangle = e^{-i\omega_c a^\dagger a t} e^{-i\epsilon_j t} |\chi_j\rangle$, $\epsilon_{1,3,4} = 0, \delta, \delta + \Delta_\mu$, and it becomes

$$i \partial_t |\chi_2\rangle = \Delta |\chi_2\rangle + \underbrace{g_0 a e^{-i\omega_c t} |\chi_1\rangle + \frac{\Omega_c}{2} e^{-i\delta t} |\chi_3\rangle}_{S(t)}, \quad (17)$$

a driven first-order equation whose source has amplitudes $g_0 \sqrt{\langle n \rangle}$, $\Omega_c/2$ and turns over at rates ω_c, δ — condition (A2), item by item. When $|\Delta|$ dominates all of them, $|\chi_2\rangle$ never builds up: it locks to the value that makes the right-hand side vanish, $|\chi_2\rangle \simeq -S(t)/\Delta$, i.e.

$$|\phi_2\rangle \simeq -\frac{1}{\Delta} \left(g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle \right). \quad (18)$$

$|2\rangle$ is no longer a degree of freedom; it is a shadow of the two states that drive it. Substituting into Eq. (14),

$$g_0 a^\dagger |\phi_2\rangle = -\frac{g_0^2}{\Delta} a^\dagger a |\phi_1\rangle - \frac{g_0 \Omega_c}{2\Delta} a^\dagger |\phi_3\rangle, \quad (19)$$

and into Eq. (16),

$$\frac{\Omega_c}{2} |\phi_2\rangle = -\frac{g_0 \Omega_c}{2\Delta} a |\phi_1\rangle - \frac{\Omega_c^2}{4\Delta} |\phi_3\rangle. \quad (20)$$

Four terms: the same four rows of the table in §4.1, and the same Eq. (6). The correspondence is exact — Eq. (18) is $|\phi_2\rangle = -\sum_n V_n^\dagger |\phi_n\rangle/\Delta$, and feeding it back multiplies by V_m .